

Stat 150– Lecture 01

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1 Markov Chain

Markov Chain: Current state is enough information

$$\mathbb{P}(X_{t+1} = a_{t+1} | X_t = a_t, X_{t-1} = a_{t-1}, \dots, X_0 = a_0) = P(X_{t+1} = a_{t+1} | X_t = a_t)$$

Notation: the **transition matrix** P has entries $P(i, j) = \mathbb{P}(X_{t+1} = j | X_t = i)$, and every row sums to 1. If μ_t is the row vector of probabilities of X_t , then $\mu_{t+1} = \mu_t P$, so $\mu_t = \mu_0 P^t$: matrix powers carry the multi-step behavior. Random walk with absorbing states is not too hard, just solve lots of equations

1.1 Reflecting Random walk on a line graph

Suppose we have a random walk X_t takes values on $S = 0, 1, 2, \dots, N$. such that

$$P(i, i+1) = p, P(i, i-1) = 1-p, P(N, N-1) = P(0, 1) = 1 \text{ and } 0 < p < 1$$

In this case, at the previously absorbing states 0 and 1, they "reflect" so we don't get stuck and with probability 1 we always return to any state

Why: the state space is finite and every state can be reached from every other state, so nothing can get stuck or escape – on a finite state space such a chain returns to each state infinitely often with probability 1 (**recurrent**; this gets made precise in the next lecture).

1.2 Infinite Line Graph

Consider the previous example except $S = \mathbb{Z}$.

Let X_t describe the position at time t of the random walk.

In this random walk, do we visit any state infinitely many times?

Answer: Yes, if and only if $p = 1 - p = \frac{1}{2}$ and the walk is symmetric.

Extension: How does this work in $\mathbb{Z}^2$ or $\mathbb{Z}^3$??

Answer to Extension: $\mathbb{Z}^2$ if symmetric, $\mathbb{Z}^3$ hell no!(transient random walk)

Intuition for both answers: if $p \neq \frac{1}{2}$ the walk *drifts* – it runs away toward $+\infty$ or $-\infty$ faster than it can come back, so every state is visited only finitely many times (**transient**). The symmetric walk $p = \frac{1}{2}$ has no drift, so it returns to every state infinitely often – though the expected time to return is infinite. The trichotomy $\mathbb{Z}/\mathbb{Z}^2/\mathbb{Z}^3$ for the symmetric walk is **Pólya's theorem**: recurrent in dimensions 1 and 2, transient in dimension ≥ 3 (Durrett computes the transience when $p \neq \frac{1}{2}$ in §1.3).

1.3 Random walk over a circle

The transition matrix is

$$P = \begin{pmatrix} 0 & p & 0 & \cdots & 0 & q \\ q & 0 & p & \ddots & \vdots & 0 \\ 0 & q & 0 & \ddots & 0 & \vdots \\ \vdots & \ddots & \ddots & \ddots & p & 0 \\ 0 & \cdots & 0 & q & 0 & p \\ p & 0 & \cdots & 0 & q & 0 \end{pmatrix}_{(N+1) \times (N+1)}, \quad p + q = 1.$$

Two things worth checking on this matrix: every *column* also sums to 1 (it is **doubly stochastic**; Durrett §1.4.1), so the uniform distribution $(\frac{1}{N+1}, \dots, \frac{1}{N+1})$ is *stationary*: $\mu P = \mu$. And with $p, q > 0$ every state reaches every other state around the circle, so the chain is irreducible.

2 Questions and follow-up

- Which of the examples in Durrett §1.1 (gambler's ruin, Ehrenfest urn, branching) are random walks in disguise?
- Why exactly does drift $p \neq \frac{1}{2}$ make the walk on $\mathbb{Z}$ transient? (Durrett §1.3 does this computation – redo it.)
- TODO: